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(Government Aided Autonomous Institute)

Department of Computer Science and Engineering

Project Synopsis

Universal Semantic Video Search & Moment Retrieval

Bachelor of Technology
in
Computer Science and Engineering

Submitted by

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1. Introduction

1.1 Background

Online and institutional video repositories have grown far faster than the tools available to search inside them. A lecture recording, a tutorial, a sports broadcast, or an interview may run for an hour or more, yet the information a user actually wants – a specific explanation, an action, or a visual detail – is often confined to a few seconds of that video. Existing video search largely relies on titles, tags, or manually written descriptions, none of which capture what is actually happening inside the video at a fine-grained, moment level. This creates a large gap between what users want to ask (“find the part where X happens”) and what current systems can answer.

1.2 Motivation

Manually scrubbing through long videos to locate a specific moment is slow and unreliable, and it does not scale as video libraries grow. A video inherently carries multiple layers of information – visual content, spoken narration, and on-screen text – and different queries depend on different combinations of these layers. A query about a spoken explanation should rely on speech; a query about a physical action should rely on visuals; a query about on-screen code or slides should rely on text recognition. No single fixed strategy serves all queries equally well, and nearest-neighbor retrieval alone will always return *some* result even when nothing relevant actually exists in the corpus. This motivates a system that adapts its retrieval strategy to the query itself and is able to recognize when it does not have a confident answer, rather than silently returning the closest – but wrong – match.

1.3 Problem Statement

Develop a multimodal system to retrieve the most relevant video and precisely localize the answer-bearing moment using adaptive visual, speech, and text evidence.

Given an arbitrary natural-language query and a heterogeneous collection of videos spanning multiple domains, retrieve the most relevant video and the precise start and end timestamp of the moment that answers the query, using a combination of visual, speech. The system must adaptively decide how much to trust each evidence type per query, verify that a genuinely relevant result exists before returning one, and spend additional computation on temporal refinement only when its initial prediction is not confident.

2. Customer Connect Details

2.1 Customer Name

2.2 Contact Person

2.3 Email / Phone

2.4 Nature of Engagement

Periodic review meetings with the project guide for problem scoping, methodology validation, and progress evaluation at each phase of the roadmap.

3. Literature Review

- [1] **Jeong et al., 2025 – “VideoRAG: Retrieval-Augmented Generation over Video Corpus.”**

Proposes a VideoRAG framework that dynamically retrieves query-relevant videos and combines visual and textual information for answer generation. It uses informative frame selection and ASR-based textual extraction when subtitles are unavailable.

Limitation: Focuses primarily on video retrieval and answer generation and does not explicitly provide corpus-level answerability detection or precise temporal moment localization.

- [2] **Ren et al., 2025 – “VideoRAG: Retrieval-Augmented Generation with Extreme Long-Context Videos.”**

Proposes a dual-channel VideoRAG framework combining graph-based textual knowledge grounding with multimodal video representations for extremely long and multiple videos. It performs query reformulation, graph-based retrieval, visual retrieval, and LLM-based filtering to identify relevant video content.

Limitation: Primarily targets long-context video understanding and RAG-based answer generation; it does not explicitly model corpus-level answerability or provide a dedicated coarse-to-fine temporal localization mechanism.

- [3] **Bai et al., 2025 – “Bridging Information Asymmetry in Text-Video Retrieval: A Data-Centric Approach.”**

Proposes a text-enrichment framework that generates event-level captions during training and diverse query variants during retrieval. A query-selection mechanism selects relevant and diverse queries to improve text-video retrieval while reducing redundant retrieval computation.

Limitation: Focuses on text-video retrieval and query enrichment rather than precise temporal moment localization, multimodal modality fusion, or corpus-level answerability detection.

- [4] **Luo et al., 2025 – “Video-RAG: Visually-aligned Retrieval-Augmented Long Video Comprehension.”**

Proposes a training-free Video-RAG pipeline that extracts visually aligned auxiliary information from videos, including ASR, OCR, and object detection. These auxiliary texts are retrieved and combined with video frames and the query to improve long-video understanding while reducing computational requirements.

Limitation: The framework is designed mainly to enhance LVLM-based long-video comprehension and does not explicitly learn query-conditioned weights across independently indexed modalities or perform corpus-level answerability detection.

- [5] **Jeon et al., 2026 – “See More, Store Less: Memory-Efficient Resolution for Video Moment Retrieval.”**

Proposes SMORE to improve memory-efficient video moment retrieval using query-guided captions, query-aware importance modulation, and structured visual compression. The method preserves important visual information while reducing redundant frames.

Limitation: The framework introduces additional processing latency and its accuracy can degrade for highly ambiguous videos or queries. It mainly focuses on memory-efficient visual retrieval rather than multimodal fusion and corpus-level answerability.

[6] **Zhao and Wang, 2026 – “ConfDiff: Confidence-Guided Representation Diffusion for Video Moment Retrieval.”**

Proposes a confidence-guided diffusion framework that models temporal boundaries as uncertain representations and selectively refines low-confidence temporal regions. The method uses a confidence map to allocate refinement effort to ambiguous parts of the video.

Limitation: Primarily addresses temporal boundary refinement after visual-text alignment and does not address corpus-level retrieval, multimodal evidence fusion across independently indexed modalities, or answerability detection.

[7] **Ao, Wang, and Boddeti, 2026 – “QSVideo: Query-Conditioned Semantic Temporal Retrieval for Video Understanding.”**

Proposes QSRanker and QSRetrieval for query-conditioned visual evidence retrieval. QSRanker reformulates questions into retrieval-friendly queries and evaluates Object, Action, and Location relevance, while QSRetrieval balances relevance, diversity, and temporal coverage.

Limitation: Primarily operates on visual evidence and does not explicitly integrate independently indexed speech and OCR modalities or determine whether a relevant answer exists in the entire corpus.

3.1 Research Gap

The reviewed works address different important aspects of video retrieval and understanding. VideoRAG introduces multimodal video retrieval and grounded answer generation, while the extreme long-context VideoRAG focuses on graph-based knowledge grounding and retrieval across long and multiple videos. The data-centric text-video retrieval approach improves query representation through query enrichment and diverse query selection. Video-RAG further introduces visually aligned auxiliary information such as ASR, OCR, and object detection for efficient long-video understanding. SMORE addresses memory-efficient video moment retrieval through query-guided captions and structured visual compression. ConfDiff focuses on uncertainty-aware temporal boundary refinement, while QSVideo improves query-conditioned visual evidence selection by balancing relevance, diversity, and temporal coverage.

However, these approaches address these capabilities largely as separate components. They do not provide a unified coarse-to-fine framework that combines **multimodal evidence indexing, query expansion, dynamic modality weighting, corpus-level answerability detection, and confidence-gated temporal refinement**.

The proposed system addresses this gap by integrating visual, speech, and OCR evidence into a common retrieval framework. It generates diverse query interpretations, dynamically determines the importance of each modality according to the query, verifies whether a relevant answer actually exists in the corpus, and performs additional temporal refinement only when the initial localization confidence is low.

Thus, the proposed work aims to provide a more reliable, explainable, and computationally efficient semantic video search system capable of returning the relevant video and precise temporal moment, while explicitly handling queries for which no confident evidence exists.

4. Objectives

1. To study existing multimodal video retrieval, video moment retrieval, VideoRAG, and multimodal fusion techniques and identify their limitations in heterogeneous video collections.
2. To investigate the effectiveness of query-conditioned dynamic weighting of visual, speech (ASR), and on-screen text (OCR) evidence compared with fixed-weight multimodal fusion.
3. To design a domain-agnostic coarse-to-fine multimodal video retrieval framework with answerability detection and adaptive query expansion for efficient retrieval.
4. To implement confidence-gated temporal localization that performs additional temporal refinement only when the initial prediction is uncertain.
5. To experimentally evaluate the proposed system across heterogeneous video domains using retrieval, temporal localization, answerability, and computational efficiency metrics.

5. Sustainable Development Goal (SDG) Mapping

This project is mapped to the following Sustainable Development Goal, in consultation with the project guide:

Mapped SDG: 9 – Industry, Innovation, and Infrastructure

Justification: The proposed project supports SDG 9 through the development of a software-based, adaptive information-retrieval infrastructure for multimodal video content. Video is one of the fastest-growing forms of digital data, and existing search infrastructure indexes videos only at the level of titles, tags, or manual descriptions, leaving the actual visual, spoken, and on-screen content effectively unsearchable at a fine-grained level. The proposed system introduces an automated, coarse-to-fine retrieval pipeline that combines multiple evidence sources – visual embeddings, speech transcripts, and on-screen text – with a novel adaptive decision layer that determines per-query modality trust, verifies whether a genuinely relevant result exists before committing further computation, and gates expensive refinement behind a confidence check. This directly supports SDG 9’s emphasis on fostering innovation and building resilient, efficient technological infrastructure, and the resulting architecture is designed to be domain-agnostic, making it a reusable and scalable piece of software infrastructure applicable across e-learning platforms, media archives, enterprise content repositories, and other organizations that need to make large video collections searchable.

6. Scope of the Project

Included in scope:

- Natural-language text query understanding, expansion, and clustering.
- Offline multimodal indexing of uploaded videos: temporal segmentation, visual embeddings (CLIP/OpenCLIP), speech transcription (Whisper), and on-screen text extraction (OCR).
- Fast approximate nearest-neighbor candidate retrieval using FAISS.
- Answerability detection to reject queries with no relevant match in the corpus.
- Dynamic, query-conditioned modality weighting for multimodal fusion.
- Temporal localization of the exact relevant moment, with confidence-gated refinement.
- A web-based demonstration interface (React + Vite) backed by a Node.js/Express application layer and a Python/FastAPI AI/ML service.
- Evaluation across at least two to three heterogeneous, publicly available video datasets.

Excluded from scope:

- Real-time processing of live video streams.
- Cross-lingual or multi-language query support (English only in this scope).
- Full diffusion-based boundary refinement (treated as optional future work, attempted only if the simpler refinement approach proves insufficient).
- Large-scale production deployment across millions of videos; the system is validated at a research/prototype scale.
- Native mobile applications.

7. Proposed Methodology

7.1 System Architecture / Approach

The proposed system is organized around three complementary approaches, each targeting a distinct stage of the retrieval problem.

1. Approach A – Offline Multimodal Indexing:

Every uploaded video is temporally segmented into short chunks. Each chunk is represented by three parallel evidence embeddings – visual (CLIP/OpenCLIP), speech transcript (Whisper), and on-screen text (OCR) – which are normalized into a common per-segment record and stored in a vector index (FAISS) alongside their metadata. This approach is intended to answer: *what evidence exists in each video, before any query is ever asked?* Because indexing is computationally heavy and only needs to happen once per video, it is treated as an asynchronous background workflow, decoupled from the online query path.

2. Approach B – Adaptive Query Understanding and Candidate Retrieval:

When a user submits a query, it is expanded into several semantically diverse variants using a large language model, then clustered and reduced to a small representative set to avoid redundant, near-duplicate retrieval passes. These representative queries are embedded and used to retrieve candidate segments from the FAISS index. This approach is intended to answer: *what is the user actually looking for, and which segments might contain it?*

3. Approach C – Confidence-Gated Multimodal Fusion and Localization:

Before any expensive multimodal computation is performed, an answerability check evaluates – using cheap signals first and escalating to multimodal checks only if the result is ambiguous – whether a genuinely relevant result exists among the retrieved candidates. If not, the system honestly reports no confident match rather than returning its closest candidate. If the query is answerable, a dynamic, query-conditioned modality-weighting mechanism fuses the visual, speech, and OCR evidence, followed by temporal localization of the exact moment. A confidence check then gates an optional refinement pass, which narrows the search window and increases temporal resolution only when the initial prediction is uncertain. This approach is intended to answer: *does a relevant answer exist at all, and if so, exactly where is it?*

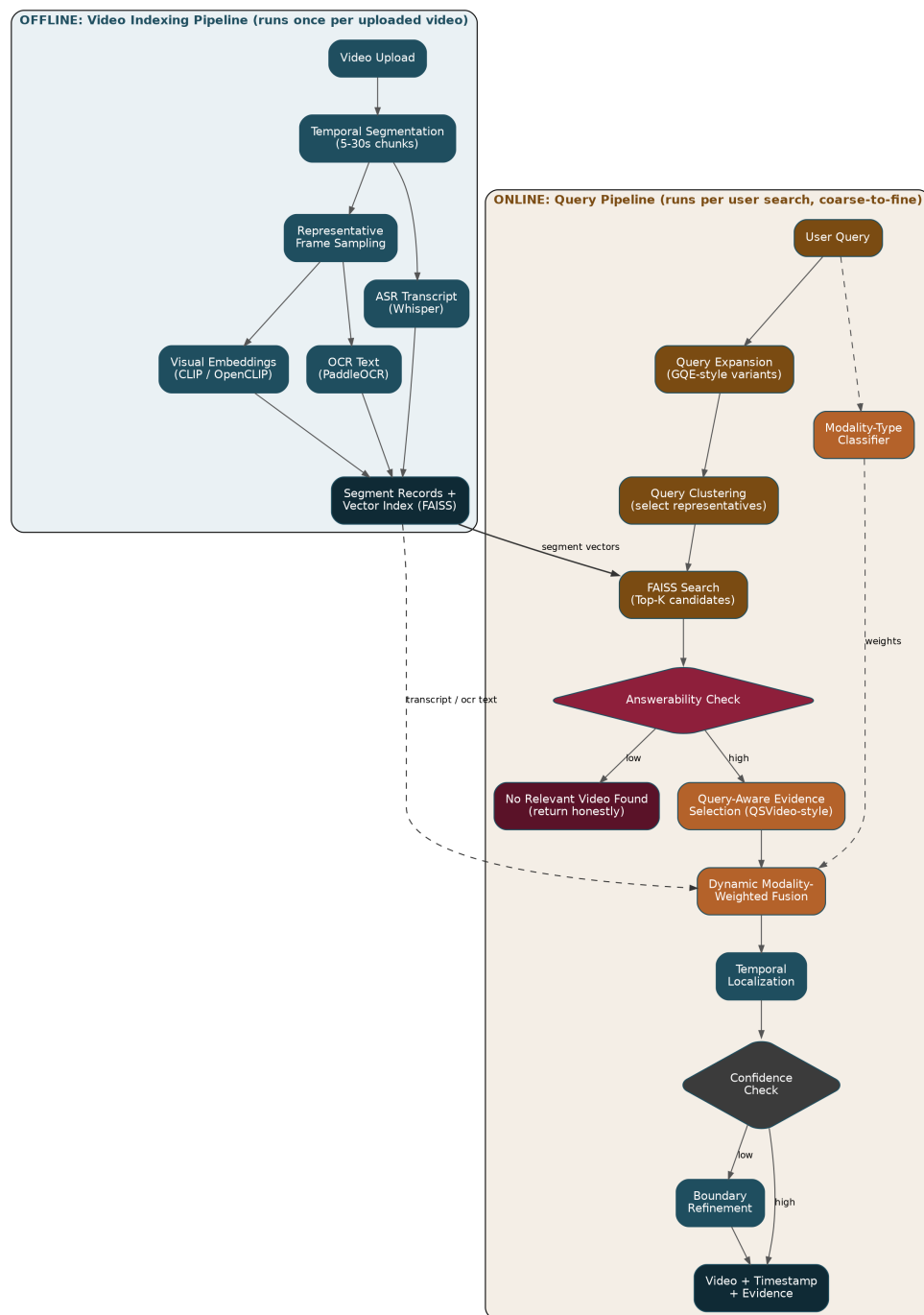


Figure 1: Proposed System Architecture: offline video indexing pipeline (left) and online query pipeline (right), including the answerability check and confidence-gated refinement branch.

7.2 Evidence Correlation and Confidence Reporting

A central part of the methodology is how evidence from the three modalities is correlated and reported rather than silently blended into a single opaque score. For every result, the system retains each modality’s individual contribution – visual similarity, speech relevance, and OCR match – together with the learned per-query weight assigned to it, so that the final ranking is explainable rather than a black box.

The system distinguishes:

1. **Visual Evidence** – similarity between the query and frame/segment-level visual embeddings.
2. **Speech Evidence** – similarity between the query and the ASR transcript of the segment.
3. **On-Screen Text Evidence** – similarity between the query and OCR-extracted text within the segment.

Where the answerability check yields a low-confidence result, the system preserves that uncertainty explicitly – either declining to return a result or clearly labelling a low-confidence candidate as such – rather than presenting an unreliable match with unwarranted certainty.

7.3 Production Strategy

Primary Query Workflow:

The online query pipeline (Approach B and C) is the primary, latency-sensitive path that a user directly interacts with: query understanding, candidate retrieval, answerability checking, fusion, and localization all occur synchronously within a single request.

Asynchronous Indexing:

Offline multimodal indexing (Approach A) runs as a background job whenever a new video is uploaded, so that expensive segmentation, embedding, ASR, and OCR extraction never block the query path.

Evidence-Based Reporting:

The final result returned to the user presents:

- the retrieved video and the predicted start/end timestamp;
- the overall relevance and answerability score;
- a per-modality evidence breakdown (visual / speech / OCR contribution);
- the localization confidence level, and whether refinement was triggered; and
- an explicit “no confident match” notice in place of a forced answer, when applicable.

This design keeps the system explainable and reduces the risk of presenting a low-confidence guess as a verified answer.

7.4 Tools and Technologies

- **Programming Language(s):** Python, JavaScript.
- **Frontend Framework:** React with Vite.
- **Application Backend:** Node.js with Express.js (REST API, authentication, request orchestration, job queuing).
- **AI/ML Backend:** Python with FastAPI, hosting all model inference (embeddings, ASR, OCR, query expansion, modality weighting, temporal localization).
- **ML Framework(s)/Library(ies):** PyTorch, Hugging Face Transformers, CLIP/OpenCLIP, Whisper, PaddleOCR/Tesseract.
- **Vector Search:** FAISS (Facebook AI Similarity Search).
- **Database:** MongoDB (video, segment, query, and result metadata).
- **Asynchronous Job Processing:** Redis with BullMQ, for offline video indexing jobs.

7.5 Work Plan / Timeline

The project is planned over an 8-week cycle for this synopsis phase, covering system design through a working and evaluated prototype.

Phase	W1	W2	W3	W4	W5	W6	W7	W8
1. Requirement Analysis & Literature Review								
2. System Design & Architecture								
3. Offline Indexing								
4. Query Expansion, Clustering, ASR/OCR								
5. Dynamic Modality Weighting & Fusion								
6. Answerability Detection & Calibration								
7. Temporal Localization & Refinement								
8. Testing, Ablations, Documentation & Demo								

Table 1: Project Work Plan and Timeline (8 Weeks)

Week 1 – Requirement Analysis, Literature Review & Dataset Selection:

Study existing video retrieval and moment-localization literature; finalize the set of evaluation datasets (e.g., QVHighlights, Charades-STA); define evidence categories and evaluation metrics.

Week 2 – System Design & Architecture Finalization:

Finalize the offline/online pipeline split, database schema, and API contracts between the frontend, application backend, and AI/ML service.

Weeks 2–3 – Offline Indexing Module:

Implement temporal segmentation, visual embedding extraction, and FAISS indexing; validate baseline visual-only retrieval.

Weeks 3–4 – Query Expansion, Clustering & ASR/OCR Integration:

Implement LLM-based query expansion with diversity-aware selection; integrate Whisper (ASR) and OCR extraction into the indexing pipeline; build a fixed-weight multimodal fusion baseline.

Weeks 4–5 – Dynamic Modality Weighting:

Implement and train the query-conditioned modality-weighting mechanism; validate it against the fixed-weight baseline.

Weeks 5–6 – Answerability Detection:

Implement the cheap-first, escalate-if-ambiguous answerability signals; calibrate the decision threshold on a labeled set of answerable/unanswerable queries.

Weeks 6–7 – Temporal Localization & Confidence-Gated Refinement:

Implement temporal boundary prediction and the confidence-gated refinement pass.

Weeks 7–8 – Testing, Ablations, Documentation & Demo:

7.6 UML Diagrams

Use-Case Diagram

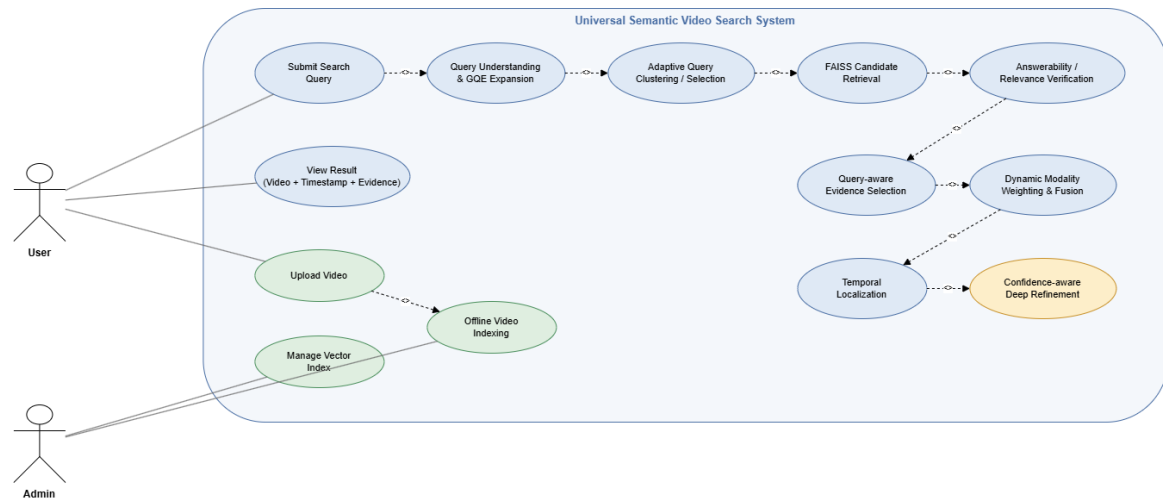


Figure 2: Use-case diagram for the Universal Semantic Video Search System

Sequence Diagram

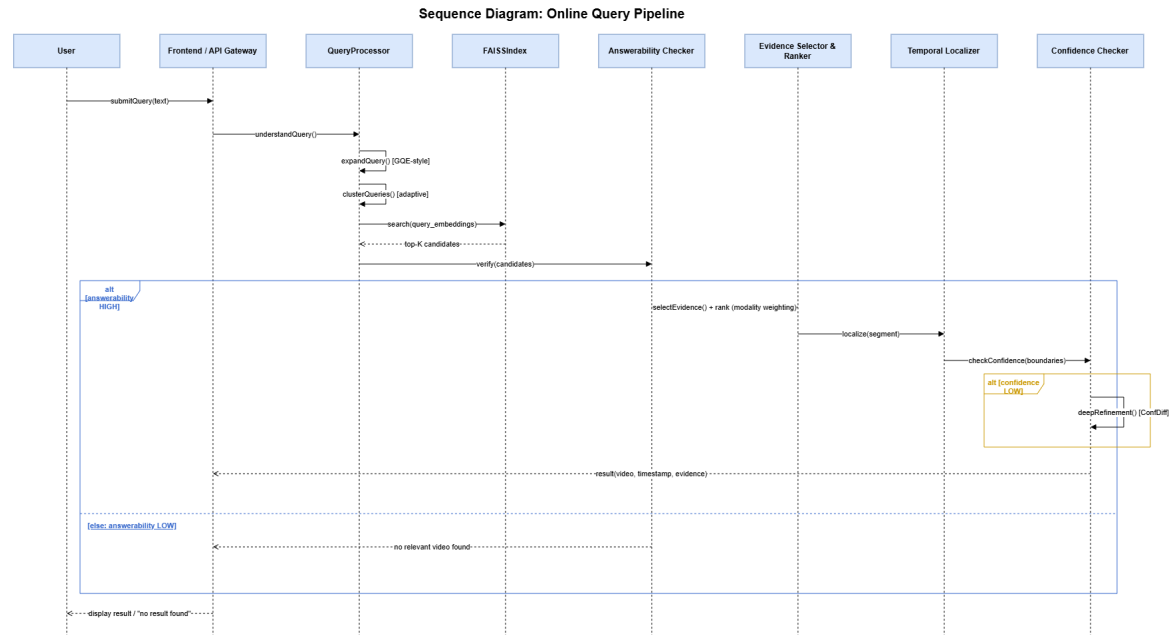


Figure 3: Sequence diagram for the online query pipeline

Activity Diagram

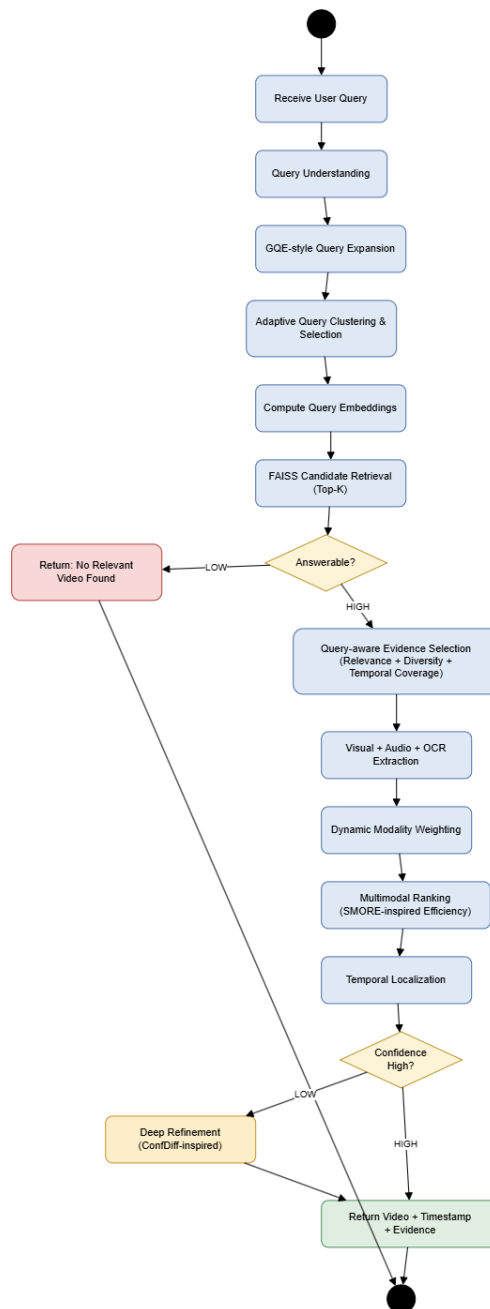


Figure 4: Activity diagram of the coarse-to-fine query pipeline

8. Expected Outcomes

- A working prototype capable of retrieving the relevant video and exact timestamp for a natural-language query across multiple, heterogeneous video domains.
- Experimental evidence, via ablation studies, of whether dynamic modality weighting improves retrieval and localization accuracy compared to fixed-weight fusion.
- A calibrated answerability detection module with measured precision, recall, and F1-score on a labeled set of answerable and deliberately unanswerable queries.
- Measured efficiency figures (query latency, percentage of queries requiring deep refinement) demonstrating that the coarse-to-fine design keeps expensive computation limited to a small fraction of queries.
- A demonstrable, web-based end-to-end system covering upload, indexing, search, and result display with supporting evidence.

9. Feasibility Study

9.1 Technical Feasibility

All core components rely on mature, well-documented, open-source tools and pretrained models (CLIP/OpenCLIP for visual embeddings, Whisper for speech transcription, PaddleOCR/Tesseract for text extraction, FAISS for vector search), removing the need to train large models from scratch. The team has prior exposure to full-stack web development (React, Node.js) and applied machine learning (Python, PyTorch), which the project directly builds on. Publicly available, pre-annotated datasets (e.g., QVHighlights, Charades-STA) further reduce the technical risk associated with data collection and annotation.

9.2 Economic Feasibility

The project uses open-source frameworks and libraries at no licensing cost. Compute requirements for indexing and evaluation can be met using free or low-cost GPU resources (e.g., academic cloud credits, Google Colab/Kaggle), and any optional LLM API usage for query expansion can be scoped to remain within a minimal, manageable budget by caching repeated queries. No specialized hardware is required for end users.

9.3 Operational Feasibility

The system is exposed through a standard web interface requiring no specialized training to use – a user simply types a query and receives a video with a timestamp and supporting evidence. The modular separation between the Node.js application layer and the Python AI/ML service allows each part to be developed, tested, and deployed independently, easing both development and eventual demonstration.

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